

Conjugacy in $SL_2(F_3)$ I

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Problem 0. Let F be a field, V a finite dim F -vector space. Note that F^* acts on $V/\{0\}$ by scaling. Let $PV := V/0/F^* =$ the quotient. Explain why PV is the set of one dimensional subspaces of V . Show that $GL(V)$ acts on PV , and on $V/0$ in such a way that the quotient map is a map of $GL(V)$ sets. Explain why $F^* =$ scalar matrices, acts trivially on PV , and so the action of $GL(V)$ on PV acts through the quotient $GL(V) \twoheadrightarrow PGL(V) := GL(V)/F^*$.

Solution. PV is the set of one dimensional subspaces in V , because it $F^* \cdot v$ consists of v scaled by F^* which is a line and is in bijection with the $span(v)$.

Let $q : V/0 \rightarrow PV$ be the quotient map. $\forall A \in GL(V)$, $q(AV) = F^* \cdot AV = \{fAv : f \in F^*\} = \{fAv : f \in F^*\} = \{Afv : f \in F^*\} = A \cdot \{fv : f \in F^*\} = A \circ F^* \cdot x = Aq(x)$.

F^* acts trivially on PV because it preserves the 1 dimensional subspace. Since $\forall F^* \cdot v \in PV$, $F^* \cdot (F^* \cdot v) = \{f_1 f_2 v : f_1 f_2 \in F^*\} = \{fv : f \in F^*\} = F^* \cdot v$ ■

We take $S := SL_2(F_3)$. $F := F_3, V := F^2$ (column vectors of size 2), F^* acts on $V/0$ by scaling, let P be the quotient, $(V/0)/F^*$ Note this is The same as the set of one dimensional subspaces of V . Let $G := GL_2(F)$

Problem 1. Show $|P| = 4, |G| = 48, |S| = 24$

Proof. $|G| = (3^2 - 3)(3^2 - 1) = 6 * 8 = 48$. $[G : S] = 2 \implies |S| = 24$ $P = \{[1 : 0], [1 : 1], [2 : 1], [0 : 1]\} \implies |P| = 4$ □

Problem 2. Explain how G acts on P . This gives a map $G \rightarrow S_4 = Perm(P)$. Show that the kernel are the scalar matrices, a copy of F^*

Proof. G acts on P by left multiplication. So $\forall g \in G, g * F^* x = F^* gx$.

Suppose $k \in \ker[G \rightarrow S_4]$, then $k * F^* e^i = F^* e^i \implies k e_i = \lambda_i e^i$. Then $k * F^* (\sum_{i=1}^n e_i) = F^* (\sum_{i=1}^n e_i) \implies k(\sum_{i=1}^n e_i) = f(\sum_{i=1}^n e_i)$. But $\sum_{i=1}^n k e_i = \sum_{i=1}^n \lambda_i e_i = (\sum_{i=1}^n f e_i) \implies \lambda_1 = \dots = \lambda_n$. Because $e_i \in V_\lambda(k)$. Thus k is a scalar matrix. Thus $\ker = F^* s$ □

Problem 3. Show the above induces an $PG \cong S_4$.

Proof. Since $\ker G \rightarrow S_4 = F^*$ and $PG := G/F^*$ then by fundamental theorem of homomorphism, $S_4 \cong F^*$. □

Problem 4. Show that the kernel of $S \rightarrow S_4$ is $F^* = \{+I, -I\}$ ($-I$ is the negative of the identity matrix)

Proof. $S \subset G$. $-I \in S$, because $\det(-I) = 1$. So if $\ker$ from $G \rightarrow S_4$ is F^* . $\ker S$ to G is $F^* \cap S = \{\pm I\}$. $\square$

Problem 5. Conclude we have an iso $PS \cong A_4$, where $PS := S/I, -I$.

Proof. $[S : PS] = 2 \implies |PS| = 24$. Also only index 2 subgroup of S_4 is A_4 this implies $PS \cong A_4$. $\square$

$Z \in Z(S)$ Let $PS := S/Z$. Now for any group G , write G/G for the set of conjugacy classes (the quotient of G for G acting on itself by conjugation). Let $q : S \rightarrow PS$ be the quotient map.

Problem 6. Explain why q induces a natural surjective map $q : S/S \rightarrow PS/PS$.

Proof. $\forall [x] \in S/S$, $q([x]) = [q(x)] = [\{zxz^{-1} : z \in Z\}]$.

$[x] = [g x g^{-1}]$. $q'([g x g^{-1}]) = [q(g x g^{-1})] = [q(g)q(x)q(g)^{-1}] = [x]$ by definition of conjugation. Thus since q is well defined it inherits the surjective property from q . $\square$

Problem 7. Show that left multiplication by Z permutes conjugacy classes, so gives a action of Z on S/S . Show that the orbits are exactly the fibres of $q : S/S \rightarrow PS/PS$

Proof. For $z \in Z$, $z \cdot [x] = \{z g x g^{-1} : g \in S\} = \{g z x g^{-1} : g \in S\} = [z x]$.

If $z \cdot [x] = z \cdot [y]$. Then $g z x g^{-1} = z y \implies z g x g^{-1} = z y \implies y \in [x]$.

$q^*([q(x)]) = q^*([qZ]) = \bigcup_{g \in G} g x Z g^{-1} = \bigcup_{g \in G} Z g x g^{-1} = Z \bigcup_{g \in G} g x g^{-1} = Z * [x]$. Thus each fibre is the orbits of left multiplication by Z . $\square$

Problem 8. Now suppose Z has prime order p . Show the inverse image under $q : S \rightarrow PS$ of a conjugacy class is either a single conjugacy class, p -times as big, or a disjoint union of p conjugacy classes, all of the same size as the original.

Proof. Since each fibre is a Z orbit. By orbit stabilizer $|Z * [x]| |Z|$. Thus $|Z * [x]| |p| \implies |Z * [x]| = 1, p$.

If $|Z * [x]| = p$, then there are p elements in the fiber of $[xZ]$ each with the same size. Then $|Z^{[x]}| = 1 \implies e \in Z$ preserves $[s]$.

When $|Z * [x]| = 1$, then there is only 1 element in the fibre of $[xZ]$ and thus there is only 1 big Conjugacy class. Since $|Z^{[x]}| = p$ then the conjugacy class is p times as big as $[xZ]$. $\square$

Problem 9. Take s in S and consider $q^*[q(s)]$ show this is either $[s]$, or it is the disjoint union of the $[zs]$ as z runs over Z . In the second case we'll say $[s]$ "breaks up". Show $[s]$ does not break up iff zs is conjugate to s for all z in Z , and this happens iff s is conjugate to zs for some non-trivial z in Z ,

Proof. Assume $|Z| = p$ where p is prime $q^*[q(s)] = Z * [s] = \bigcap_{z \in Z} [zs]$.

If zs is conjugate to s , thus $zs = g s g^{-1}$ then $z * zs = z^2 s = z g s g^{-1} = g z s g^{-1}$ thus $z^2 s$ is conjugate to zs . Thus $z^k s$ is conjugate to s . Thus $[s] = [zs] = \dots = [z^{(p-1)} s]$. Thus $q^*[q(s)] = [s]$

Thus iff $\exists z \in Z \ni zs \sim s \implies z^k s \sim s \iff [s]$ doesn't break up.

Otherwise if zs is not conjugate to s , by the previous problem it breaks into p conjugacy classes. $\square$